

What You Should Know When Running & Studying `comp40771_lab10.ipynb`

The Jupyter notebook is designed to let you **test each major component of the Lab 10 system**:

- MongoDB (database)
- Ollama (LLM for reasoning)
- Sandbox API (execution environment)
- Telemetry pipeline (monitoring signals)

Even though your Ollama container is misbehaving, the notebook itself is still structured for learning and testing — *you can still understand and study every part*.

Below is the simplified breakdown of the notebook's logic.

1. Dependency Setup

You'll see code like:

Python

```
!pip install pymongo==4.10.1
```

Show more lines

This part ensures that the container has the required Python libraries.

👉 **You only need to run this if imports fail.**

Otherwise skip it.

2. MongoDB Configuration

The notebook loads environment variables:

Python

```
MONGO_URI = os.getenv('MONGO_URI', 'mongodb://mongodb:27017')  
MONGO_DB = os.getenv('MONGO_DB', 'lab10_audit')
```

Show more lines

What to understand:

- The notebook connects to the MongoDB container via the hostname `mongodb`
- `lab10_audit` is the database where all results will be stored
- These names come from your `docker-compose.yml`

This teaches you how multiple containers communicate in a Docker network.

3. Connecting to MongoDB

Python

```
client = MongoClient(MONGO_URI)  
db = client[MONGO_DB]  
print(db.list_collection_names())
```

Show more lines

Important concept:

- This shows you how to interact with MongoDB programmatically.
 - You learn how collections work.
 - You see which collections already exist.
-

4. Creating Collections & Indexes

The notebook creates collections like:

- `sandbox_runs`
- `telemetry_logs`
- `sandbox_classifications`
- `reports`

and ensures they have indexes (for faster queries).

What you learn:

- How MongoDB organizes data.
 - How a good audit database is structured.
 - Why indexes matter for querying performance.
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5. Inserting Test Documents

The notebook inserts example data:

- A fake run document
- Fake telemetry
- Fake classification
- A fake final report

Why this matters:

This section shows you the **shape of the data** your actual system will produce later:

JSON

```
{  
  "run_id": "run_001",  
  "api_id": "api07",  
  "classification": "suspicious",  
  ...  
}
```

Show more lines

The notebook demonstrates:

- inserting,
 - updating,
 - deleting,
 - and querying documents.
-

6. Querying MongoDB

You learn simple analytics:

- Find a run by `run_id`
- List all suspicious classifications
- Count all documents in each collection

 **This teaches you:**

- How to inspect stored evidence
 - How to validate that data is being logged correctly
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7. Testing Ollama (LLM Reasoning)

This part tries:

Python

```
requests.get("http://ollama:11434/")
```

Show more lines

and:

Python

```
requests.post(f"{OLLAMA_BASE}/api/generate", json=payload)
```

Show more lines

 **Purpose:**

- Verify LLM is running
- Query a model
- Test text generation

👉 It is *okay* that this part errors out right now —
you can still learn *what it is supposed to do*.

8. Testing the Sandbox API

This is VERY important.

The notebook interacts with:

- `/status`
- `/telemetry/start`
- `/execute`
- `/telemetry/status`
- `/results/last`

 **This teaches you the workflow:**

1. Start telemetry
2. Run a sandbox test
3. Poll telemetry endpoints
4. Store retrieved evidence
5. Produce explanations later

This section is crucial because **you will implement a similar pipeline** in your main project.

9. Full Investigation Workflow Pattern

The notebook's sandbox code demonstrates the main idea:

- **Send request → Get run_id → Monitor behaviour → Get classification**

Your final system will:

- correlate sandbox classifications with your own,
 - store them in MongoDB,
 - and produce a transparent explanation.
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10. What the notebook does NOT do

The notebook does **not**:

- build the full reasoning pipeline,
- run CI probing,
- orchestrate agents,
- or generate full reports.

Instead, it gives you the **tools** to test each piece before combining them.

In Summary — What You Should Know

Here is the simplest possible breakdown:

✓ MongoDB section

- Teaches you how the audit database works
- How to store & retrieve investigation data

✓ Ollama section

- Teaches how reasoning (explanations) will be generated
- But ok if it fails for now

✓ Sandbox section

- Teaches the telemetry workflow
- Teaches how the external sandbox API behaves

✓ Everything together

- Helps you understand how to build the full Lab 10 system.